

PHARMACEUTICAL DOSAGE FORMS

Tablet Formulation

Process · Excipients · Stability · Challenges

Dosage Form Selected: Compressed Tablet (Immediate Release)

Pharmaceutical Sciences | Formulation & Development

TABLET: OVERVIEW & TYPES

A tablet is a solid unit dosage form produced by compressing a drug with excipients. It is the most widely used oral dosage form due to accuracy, stability, and ease of manufacture.

Immediate Release (IR)

Drug released rapidly post-ingestion. Most common.

Extended Release (ER)

Controlled drug release over hours. Reduces dosing frequency.

Chewable Tablet

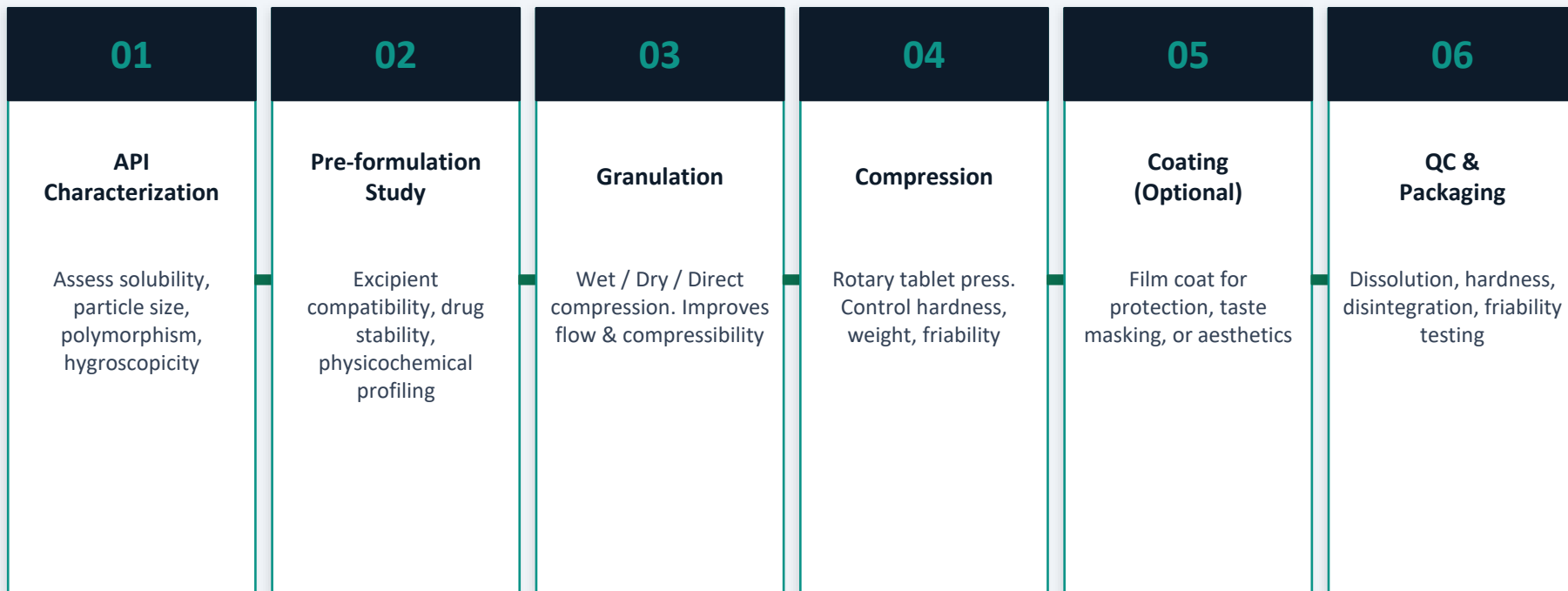
Soft texture, flavored. For pediatrics & antacids.

Dispersible Tablet

Disintegrates in water before administration.

Focus of this presentation: Immediate Release Compressed Tablet — the reference standard for formulation science.

TABLET FORMULATION PROCESS



Each step requires rigorous process validation before scale-up to commercial manufacturing.

EXCIPIENTS USED IN TABLET FORMULATION

Excipients are inactive ingredients that aid manufacturing, stability, and drug release.

Excipient Category	Function	Common Examples	Typical % w/w
Diluent / Filler	Adds bulk to low-dose API	Lactose, MCC (Avicel), Mannitol	30 – 80%
Binder	Holds granules together	PVP (Kollidon), HPMC, Starch	2 – 10%
Disintegrant	Promotes tablet breakup in GI tract	Croscarmellose Na, SSG, Starch	2 – 8%
Lubricant	Prevents sticking to punch/die	Magnesium Stearate, Stearic Acid	0.25 – 1%
Glidant	Improves powder flow	Colloidal SiO ₂ (Aerosil), Talc	0.1 – 0.5%
Coating Agent	Film coat: protection / appearance	HPMC, Opadry, Eudragit	2 – 5% weight gain

STABILITY FACTORS

PHYSICAL STABILITY

Moisture / Humidity

Causes softening, sticking, disintegration failure. Control with desiccants.

Temperature

Elevated temp accelerates polymorphic transitions and melting of lubricants.

Light

Photodegradation of light-sensitive APIs (e.g., nifedipine). Use amber packaging.

Mechanical Stress

Poor hardness → friability, capping, lamination during transport.

CHEMICAL STABILITY

Hydrolysis

Water cleaves ester/amide bonds in API. Moisture control is critical.

Oxidation

Free-radical degradation. Use antioxidants (BHA, BHT) or nitrogen purge.

Drug-Excipient Interaction

Maillard reaction: amine APIs + lactose → colored products.

Microbial Degradation

Moisture + organic excipients → microbial growth. Control $A_w < 0.6$.

KEY FORMULATION CHALLENGES

01

Poor Solubility (BCS Class II & IV)

- ▶ ~40% of new APIs have low aqueous solubility
- ▶ Solutions: Solid dispersions, nanosizing, cyclodextrins, lipid-based systems
- ▶ Challenge affects both dissolution rate and bioavailability

02

Poor Flowability & Compressibility

- ▶ Fine API particles have poor flow → weight variation in tablets
- ▶ Solutions: Wet granulation, slugging (dry granulation), spray drying
- ▶ Direct compression preferred but needs good flow excipients

03

Content Uniformity for Low-Dose Drugs

- ▶ API < 5% of tablet weight is critical — blend uniformity essential
- ▶ Risk of dose dumping or sub-therapeutic levels
- ▶ Solutions: Geometric dilution, high-shear mixing, ordered mixing

04

Scale-Up from Lab to Commercial

- ▶ Process parameters change non-linearly on scale-up
- ▶ Granule properties, compaction force, coating uniformity all shift
- ▶ Risk: dissolution profile changes even with same formulation

QUALITY CONTROL TESTS FOR TABLETS

IN-PROCESS CONTROLS

Test	Acceptance Criterion	Instrument
Hardness	4–10 kPa (depends on tablet size)	Tablet hardness tester
Thickness	Consistent ± 0.1 mm	Micrometer / online sensor
Weight Variation	$\pm 5\%$ of average weight (USP)	Analytical balance
Friability	< 1% weight loss after 100 rotations	Roche friabilator

FINISHED PRODUCT TESTS

Test	Specification	Method
Disintegration	< 15 min (IR tablets)	Disintegration tester
Dissolution	$\geq 80\%$ in 30 min (Q30)	USP Apparatus I or II
Content Uniformity	85–115% of label claim	HPLC / UV assay
Assay	98.0–102.0% of declared strength	HPLC

Regulatory Compliance Note

All tablet QC methods must be validated per ICH Q2(R1). Stability studies follow ICH Q1A(R2). Dissolution specifications are set based on bioequivalence data and pharmacopeial requirements (USP/BP/Ph.Eur).

Tablet Formulation: Summary

01

Formulation is multi-step —

From API characterization → granulation → compression → coating → QC, each step is interdependent.

02

Excipients are not inert —

They directly impact solubility, stability, release rate, and manufacturability. Selection is science-driven.

03

Stability = ICH compliance —

Physical + chemical stability must be evaluated under real-time and accelerated conditions per ICH Q1A.

04

Challenges are API-specific —

Poor solubility, flow, and content uniformity are the top 3 barriers — each requiring a tailored approach.

05

Scale-up needs validation —

Lab-to-commercial transition is high-risk. Process analytical technology (PAT) helps bridge the gap.